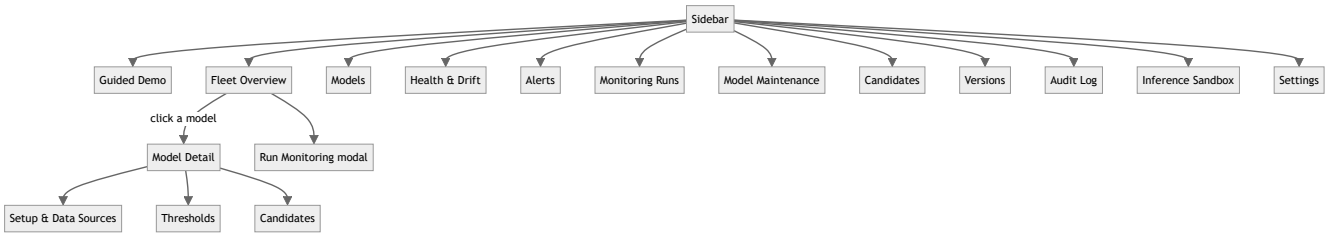
AI MODEL HEALTH MONITORING & AUTOMATED MAINTENANCE SYSTEM

Front End Requirement Document

Draft · 2026-09-29

React + Vite single page app. Dark theme. Styles in `index.css`. All backend calls go through `frontend/src/api.js`. Navigation is a sidebar (`Sidebar.jsx`) plus top bar (`Header.jsx`); `App.jsx` holds the selected tab and unit.

1. Screen map



2. Screens

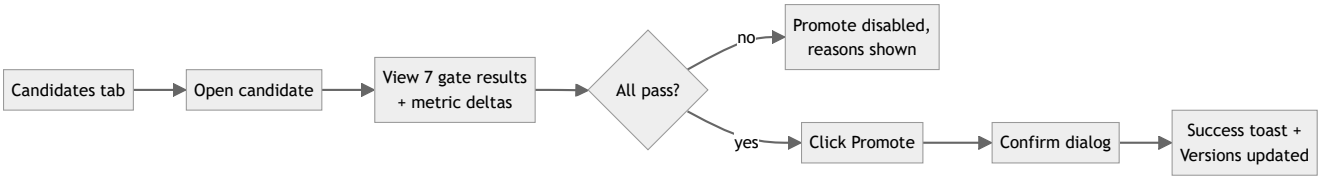
Screen	File	User can
Fleet Overview	`FleetDashboardView`	See all models with health cards; create a unit; start a run
Model Detail	`ModelDetailView`	See 5 pillar meters, candidates, thresholds, log
Setup & Data	`ModelSetupView`	Upload champion, baseline, datasets; check schema; map columns by hand
Health & Drift	`HealthView`	Read metrics, drift table, gauges
Alerts	`AlertsView`	Filter, acknowledge, resolve
Monitoring Runs	`PipelineView`, `RunMonitoringModal`	Step-by-step run with live status
Model Maintenance	`DiagnosisView`	Plain-English cause, feature-importance plots, suggested fix
Candidates	`RefinementView`	Start refinement; see each gate pass/fail
Versions	`VersionsView`	See history; roll back
Audit Log	`AuditLogView`	Read who/what/when
Inference Sandbox	`InferenceView`	Try predictions by hand
Settings	`SettingsView`	Change preferences
Guided Demo	`GuidedDemoView`	Follow a scripted end-to-end tour

Note: `IntakeView` was deleted (uncommitted); intake is now part of `ModelSetupView`.

3. UI rules

1. **Status colours:** Healthy = green, Warning = amber, Critical = red, Unknown/Delayed = grey. Never colour alone; always add a label.
2. **Plain language first:** each finding shows a short sentence, technical numbers beneath.
3. **Risky actions** (promote, rollback, delete, reset) need a confirm step and show what will change.
4. **Every screen** needs loading, empty, and error states. Errors show the server's safe message.
5. **Unit context:** the selected model is always visible in the header.
6. **Missing labels:** show "Labels delayed/unavailable", never a made-up score.
7. **Uploads:** show allowed types and the 50 MB limit before upload; show progress.
8. **Alerts badge** in the sidebar shows open count.
9. **Responsive:** usable at laptop width; tables scroll, not overflow the page.
10. **Accessibility:** keyboard reachable buttons, visible focus, text contrast on dark theme.

4. Key screen flow: promote a candidate



5. Front end gaps

- Send the API token from `api.js`.
- Hide or disable actions by role once roles exist.
- The largest components (`ModelSetupView` ~1900 lines, `FleetDashboardView`, `GuidedDemoView`) should be split.
- No automated front end tests yet.